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CSD-1900-4618 - Introducción a Ciberseguridad

11/25/2025

Proyecto Final:

Seguridad a un Entorno Simulado

Introducción:

Para la demostración del entorno simulado, usé tres máquinas virtuales emuladas por el software Oracle VirtualBox. Los sistemas operativos que elegí fueron dos distribuciones de Linux: el SO Ubuntu 22.04 (al que llamaré como "víctima"), le los diferentes servicios para recibir los ataques sin defensa. El otro SO es Kali Linux (al que llamare como el "atacante") para usar las herramientas de ataque o simulación y, por último, una copia de la víctima (al que nombre como "tortuga") con los mismos servicios, pero con otra configuración.

Los servicios básicos instalados a la MV víctima fueron los establecidos del proyecto: Servicios de Red: SSH, FTP/SFTP, DNS (Bind9, dnsmasq), DHCP, HTTP/HTTPS (Apache, Nginx).

Servicios de Base de Datos: MySQL/MariaDB, PostgreSQL.

Servicios de Correo: SMTP (Postfix, Sendmail), IMAP/POP3 (Dovecot).

Servicios de Autenticación y Control: LDAP (OpenLDAP), Kerberos, RADIUS.

Servicios de Monitoreo y Logs: Syslog/rsyslog, SNMP, Nagios/Zabbix/Prometheus.

Servicios de Seguridad y Protección: Firewall (ufw, iptables), Fail2ban.

Las herramientas de ataque que se utilizaron venían instaladas en la MV del atacante:

Nmap para la detección de puertos y servicios.

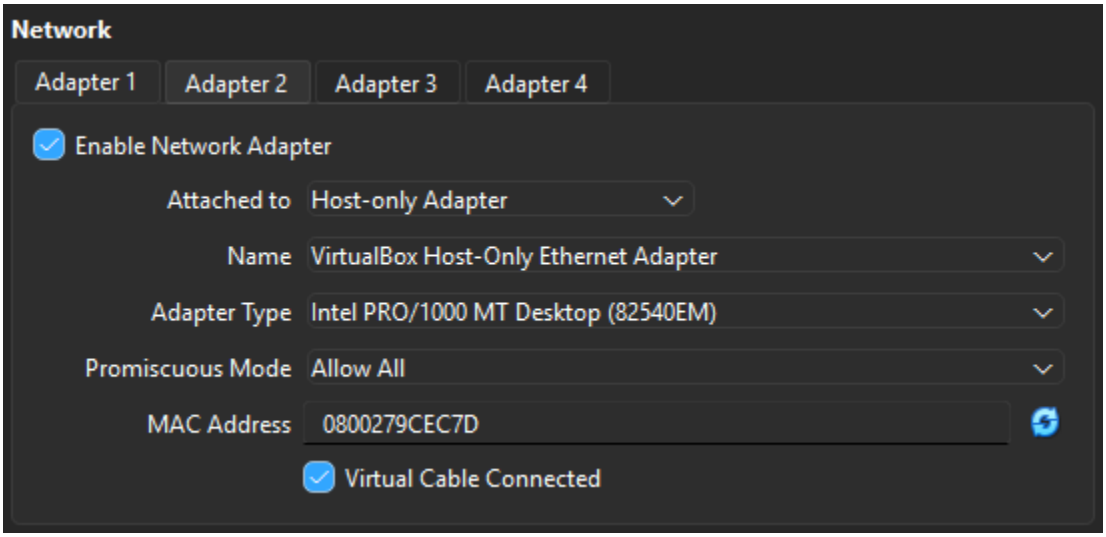
Hydra para realizar pruebas de contraseñas débiles mediante ataques de fuerza bruta.

Wireshark para la captura y análisis de paquetes de red.

I - Configuración del Entorno, comandos básicos y Servicios Instalados.

Empezando con la víctima, lo primero que hice fue configurar la VM de Oracle correctamente para dirigir el ataque, luego, dentro de la víctima, configure el SO inicial a mi gusto. Cuando me dejo manipularlo, ejecute comandos básicos para orientarme y comprobar que el sistema funcionaba bien. Verifiqué el usuario con **whoami**, y confirmé la IP usando **ip a**. Utilicé **sudo** que es que da los permisos de administrador antes de cada comando. Actualicé el sistema con **sudo apt update** y **sudo apt upgrade -y.** Me moví entre carpetas usando **ls y cd**. También revisé procesos y servicios con **systemctl status**. Instalé los servicios con **sudo apt install** y modifique los archivos de los servicios con **sudo nano /etc/ (nombre del archivo del servicio**) también desactive el firewall que traía ya instalado con **sudo ufw disable** para un ataque más efectivo.

Configuración de la MV:



Comencé instalando los diferentes servicios de red, empezando por:

el ssh

```
  Swap usage:              0%
  Processes:               121
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.0.62
  IPv6 address for enp0s3: 2001:1308:268d:3700:a00:27ff:fe9c:ec7d

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

57 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Fri Nov 21 04:50:19 UTC 2025 on tty1
ricardo@ubuntu:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 18:56:36 UTC; 13min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 654 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 707 (sshd)
      Tasks: 1 (limit: 2219)
     Memory: 3.0M
        CPU: 51ms
     CGroup: /system.slice/ssh.service
             └─707 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Warning: some journal files were not opened due to insufficient permissions.
ricardo@ubuntu:~$
```

Comandos utilizados:

Instalar: sudo apt install openssh-server

Editar configuración: sudo nano /etc/ssh/sshd_config

FTP / SFTP

```
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem       sftp    /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#       X11Forwarding no
#       AllowTcpForwarding no
#       PermitTTY no
#       ForceCommand cvs server



ricardo@ubuntu:~$ sudo sshd -t
ricardo@ubuntu:~$ systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/lib/systemd/system/vsftpd.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 18:56:36 UTC; 21min ago
    Process: 663 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 668 (vsftpd)
      Tasks: 1 (limit: 2219)
     Memory: 2.1M
        CPU: 14ms
     CGroup: /system.slice/vsftpd.service
             └─668 /usr/sbin/vsftpd /etc/vsftpd.conf

Warning: some journal files were not opened due to insufficient permissions.
ricardo@ubuntu:~$ _
```

Instalar: sudo apt install vsftpd

Editar configuración: sudo nano /etc/vsftpd.conf

DNS (Bind9)

```
#       ForceCommand cvs server



ricardo@ubuntu:~$ sudo sshd -t
ricardo@ubuntu:~$ systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/lib/systemd/system/vsftpd.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 18:56:36 UTC; 21min ago
    Process: 663 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 668 (vsftpd)
      Tasks: 1 (limit: 2219)
     Memory: 2.1M
        CPU: 14ms
     CGroup: /system.slice/vsftpd.service
             └─668 /usr/sbin/vsftpd /etc/vsftpd.conf

Warning: some journal files were not opened due to insufficient permissions.
ricardo@ubuntu:~$ grep -1 sftp /etc/ssh/sshd_config
# override default of no subsystems
Subsystem       sftp    /usr/lib/openssh/sftp-server

ricardo@ubuntu:~$ systemctl status bind9
● named.service - BIND Domain Name Server
     Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 18:56:37 UTC; 24min ago
       Docs: man:named(8)
    Process: 645 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0/SUCCESS)
   Main PID: 689 (named)
      Tasks: 6 (limit: 2219)
     Memory: 31.0M
        CPU: 236ms
     CGroup: /system.slice/named.service
             └─689 /usr/sbin/named -u bind

Warning: some journal files were not opened due to insufficient permissions.
ricardo@ubuntu:~$
```

Instalar: sudo apt install bind9

Editar configuración: sudo nano /etc/bind/named.conf.options

DHCP

```
       Docs: man:named(8)
    Process: 660 ExecStart=/usr/sbin/named $OPTIONS (code=exited, status=0/SUCCESS)
   Main PID: 743 (named)
      Tasks: 6 (limit: 2219)
     Memory: 28.5M
        CPU: 217ms
     CGroup: /system.slice/named.service
             └─743 /usr/sbin/named -u bind

Warning: some journal files were not opened due to insufficient permissions.

● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2025-11-25 11:37:22 UTC; 1min 53s ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 777 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 877 (apache2)
      Tasks: 55 (limit: 2219)
     Memory: 7.4M
        CPU: 138ms
     CGroup: /system.slice/apache2.service
             ├─877 /usr/sbin/apache2 -k start
             ├─878 /usr/sbin/apache2 -k start
             └─879 /usr/sbin/apache2 -k start

Warning: some journal files were not opened due to insufficient permissions.
ricardo@ubuntu:~$ systemctl status isc-dhcp-server
× isc-dhcp-server.service - ISC DHCP IPv4 server
     Loaded: loaded (/lib/systemd/system/isc-dhcp-server.service; enabled; vendor preset: enabled)
     Active: failed (Result: exit-code) since Tue 2025-11-25 11:37:21 UTC; 2min 45s ago
       Docs: man:dhcpd(8)
    Process: 653 ExecStart=/bin/sh -ec      CONFIG_FILE=/etc/dhcp/dhcpd.conf;      if [ -f /etc/lts>
   Main PID: 653 (code=exited, status=1/FAILURE)
        CPU: 34ms

Warning: some journal files were not opened due to insufficient permissions.
lines 1-9/9 (END)
```

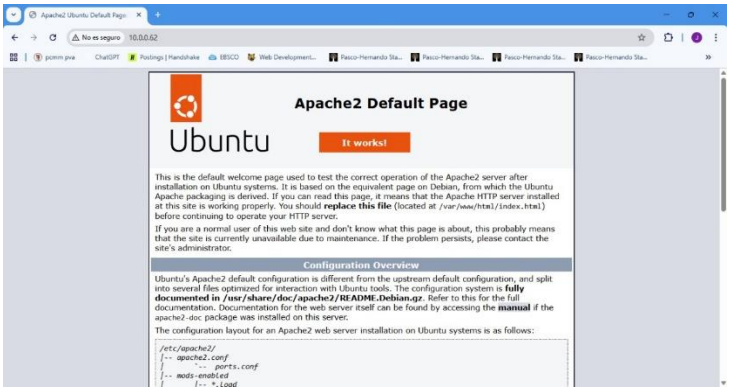
Instalar: sudo apt install isc-dhcp-server

Editar configuración: sudo nano /etc/dhcp/dhcpd.conf

HTTP/HTTPS (Apache)

```
ricardo@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:9c:ec:7d brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.62/24 metric 100 brd 10.0.0.255 scope global dynamic enp0s3
       valid_lft 256211sec preferred_lft 256211sec
    inet6 2001:1308:268d:3700:a00:27ff:fe9c:ec7d/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 85965sec preferred_lft 85965sec
    inet6 fe80::a00:27ff:fe9c:ec7d/64 scope link
       valid_lft forever preferred_lft forever
ricardo@ubuntu:~$
ricardo@ubuntu:~$
ricardo@ubuntu:~$
ricardo@ubuntu:~$
ricardo@ubuntu:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 18:56:37 UTC; 54min ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 755 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 974 (apache2)
      Tasks: 55 (limit: 2219)
     Memory: 7.4M
        CPU: 850ms
     CGroup: /system.slice/apache2.service
             ├─974 /usr/sbin/apache2 -k start
             ├─1040 /usr/sbin/apache2 -k start
             └─1041 /usr/sbin/apache2 -k start

Warning: some journal files were not opened due to insufficient permissions.
ricardo@ubuntu:~$ _
```



Instalar: sudo apt install apache2

Editar configuración: sudo nano /etc/apache2/apache2.conf

MySQL / MariaDB

```
l.service.
Setting up libcgi-pm-perl (4.54-1) ...
Setting up libhtml-template-perl (2.97-1.1) ...
Setting up mysql-server (8.0.44-0ubuntu0.22.04.1) ...
Setting up libcgi-fast-perl (1:2.15-1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.11) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ricardo@ubuntu:~$
ricardo@ubuntu:~$ systemctl status mysqul
Unit mysqul.service could not be found.
ricardo@ubuntu:~$ systemctl status mysql
● mysql.service - MySQL Community Server
     Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 20:11:56 UTC; 46s ago
    Process: 1928 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCC
   Main PID: 1936 (mysqld)
     Status: "Server is operational"
      Tasks: 38 (limit: 2219)
     Memory: 364.1M
        CPU: 13.433s
     CGroup: /system.slice/mysql.service
             └─1936 /usr/sbin/mysqld
lines 1-11/11 (END)
ricardo@ubuntu:~$
ricardo@ubuntu:~$
```

Instalar: sudo apt install mysql-server

Editar configuración: sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf

PostgreSQL

```
fixing permissions on existing directory /var/lib/postgresql/14/main ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default max_connections ... 100
selecting default shared_buffers ... 128MB
selecting default time zone ... Etc/UTC
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
update-alternatives: using /usr/share/postgresql/14/man/man1/postmaster.1.gz to provide /usr/share/m
an/man1/postmaster.1.gz (postmaster.1.gz) in auto mode
Setting up postgresql-contrib (14+238) ...
Setting up postgresql (14+238) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.11) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ricardo@ubuntu:~$ systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
     Loaded: loaded (/lib/systemd/system/postgresql.service; enabled; vendor preset: enabled)
     Active: active (exited) since Sun 2025-11-23 20:16:37 UTC; 25s ago
    Process: 4251 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 4251 (code=exited, status=0/SUCCESS)
        CPU: 7ms
ricardo@ubuntu:~$
```

Instalar: sudo apt install postgresql postgresql-contrib

Editar configuración: sudo nano /etc/postgresql/14/main/postgresql.conf

SMTP (Postfix)

```
setting inet_protocols: all
/etc/aliases does not exist, creating it.
WARNING: /etc/aliases exists, but does not have a root alias.

Postfix (main.cf) is now set up with a default configuration.  If you need to
make changes, edit /etc/postfix/main.cf (and others) as needed.  To view
Postfix configuration values, see postconf(1).

After modifying main.cf, be sure to run 'systemctl reload postfix'.

Running newaliases
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /lib/systemd/system/po
stfix.service.
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for rsyslog (8.2112.0-2ubuntu2.2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ricardo@ubuntu:~$ systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
     Loaded: loaded (/lib/systemd/system/postfix.service; enabled; vendor preset: enabled)
     Active: active (exited) since Sun 2025-11-23 20:30:32 UTC; 1min 29s ago
       Docs: man:postfix(1)
    Process: 2368 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 2368 (code=exited, status=0/SUCCESS)
        CPU: 2ms
ricardo@ubuntu:~$
```

Instalar: sudo apt install postfix

Editar configuración: sudo nano /etc/postfix/main.cf

IMAP / POP3 (Dovecot)

```
Creating config file /etc/dovecot/conf.d/20-imap.conf with new version
Setting up dovecot-pop3d (1:2.3.16+dfsg1-3ubuntu2.6) ...

Creating config file /etc/dovecot/conf.d/20-pop3.conf with new version
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.11) ...
Processing triggers for dovecot-core (1:2.3.16+dfsg1-3ubuntu2.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ricardo@ubuntu:~$ systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
     Loaded: loaded (/lib/systemd/system/dovecot.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 20:34:21 UTC; 26s ago
       Docs: man:dovecot(1)
             https://doc.dovecot.org/
   Main PID: 5912 (dovecot)
     Status: "v2.3.16 (7e2e900c1a) running"
      Tasks: 4 (limit: 2219)
     Memory: 3.2M
        CPU: 112ms
     CGroup: /system.slice/dovecot.service
             ├─5912 /usr/sbin/dovecot -F
             ├─5914 dovecot/anvil
             ├─5915 dovecot/log
             └─5916 dovecot/config
ricardo@ubuntu:~$ _
```

Instalar: sudo apt install dovecot-core dovecot-imapd dovecot-pop3d

Editar configuración: sudo nano /etc/dovecot/dovecot.conf

LDAP (OpenLDAP)

```
 systemctl restart packagekit.service postgresql@14-main.service
^[[B
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ricardo@ubuntu:~$ systemctl status sldap
Unit sldap.service could not be found.
ricardo@ubuntu:~$ systemctl status slapd
● slapd.service - LSB: OpenLDAP standalone server (Lightweight Directory Access Protocol)
     Loaded: loaded (/etc/init.d/slapd; generated)
    Drop-In: /usr/lib/systemd/system/slapd.service.d
             └─slapd-remain-after-exit.conf
     Active: active (running) since Sun 2025-11-23 20:41:47 UTC; 2min 47s ago
       Docs: man:systemd-sysv-generator(8)
    Process: 6324 ExecStart=/etc/init.d/slapd start (code=exited, status=0/SUCCESS)
      Tasks: 3 (limit: 2219)
     Memory: 3.3M
        CPU: 41ms
     CGroup: /system.slice/slapd.service
             └─6331 /usr/sbin/slapd -h "ldap:/// ldapi:///" -g openldap -u openldap -F /etc/ldap/sl>
lines 1-12/12 (END)
```

Instalar: sudo apt install slapd ldap-utils

Editar configuración: sudo nano /etc/ldap/ldap.conf

Kerberos

```
Now that your realm is set up you may wish to create an administrative
principal using the addprinc subcommand of the kadmin.local program.
Then, this principal can be added to /etc/krb5kdc/kadm5.acl so that
you can use the kadmin program on other computers.  Kerberos admin
principals usually belong to a single user and end in /admin.  For
example, if jruser is a Kerberos administrator, then in addition to
the normal jruser principal, a jruser/admin principal should be
created.

Don't forget to set up DNS information so your clients can find your
KDC and admin servers.  Doing so is documented in the administration
guide.
ricardo@ubuntu:~$ sudo systemctl status krb5-kdc
● krb5-kdc.service - Kerberos 5 Key Distribution Center
     Loaded: loaded (/lib/systemd/system/krb5-kdc.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 20:52:48 UTC; 7s ago
    Process: 7194 ExecStart=/usr/sbin/krb5kdc -P /var/run/krb5-kdc.pid $DAEMON_ARGS (code=exited, s
   Main PID: 7195 (krb5kdc)
      Tasks: 1 (limit: 2219)
     Memory: 1.1M
        CPU: 85ms
     CGroup: /system.slice/krb5-kdc.service
             └─7195 /usr/sbin/krb5kdc -P /var/run/krb5-kdc.pid

Nov 23 20:52:48 ubuntu krb5kdc[7194]: Setting pktinfo on socket 0.0.0.0.88
Nov 23 20:52:48 ubuntu krb5kdc[7194]: Setting up UDP socket for address ::.88
Nov 23 20:52:48 ubuntu krb5kdc[7194]: setsockopt(12,IPV6_V6ONLY,1) worked
Nov 23 20:52:48 ubuntu systemd[1]: Started Kerberos 5 Key Distribution Center.
Nov 23 20:52:48 ubuntu krb5kdc[7194]: Setting pktinfo on socket ::.88
Nov 23 20:52:48 ubuntu krb5kdc[7194]: Setting up TCP socket for address 0.0.0.0.88
Nov 23 20:52:48 ubuntu krb5kdc[7194]: Setting up TCP socket for address ::.88
Nov 23 20:52:48 ubuntu krb5kdc[7194]: setsockopt(14,IPV6_V6ONLY,1) worked
Nov 23 20:52:48 ubuntu krb5kdc[7194]: set up 6 sockets
Nov 23 20:52:48 ubuntu krb5kdc[7195]: commencing operation
lines 1-21/21 (END)
```

Instalar: sudo apt install krb5-kdc krb5-admin-server

Editar configuración: sudo nano /etc/krb5kdc/kdc.conf

RADIUS (FreeRADIUS)

```
Setting up freetds-common (1.3.6-1) ...
Setting up libwbclient0:amd64 (2:4.15.13+dfsg-0ubuntu1.10) ...
Setting up libct4:amd64 (1.3.6-1) ...
Setting up freeradius (3.0.26~dfsg~git20220223.1.00ed0241fa-0ubuntu3.4) ...
update-rc.d: warning: start and stop actions are no longer supported; falling back to defaults
Created symlink /etc/systemd/system/multi-user.target.wants/freeradius.service → /lib/systemd/system
/freeradius.service.
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.11) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ricardo@ubuntu:~$ systemctl status freeradius
● freeradius.service - FreeRADIUS multi-protocol policy server
     Loaded: loaded (/lib/systemd/system/freeradius.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 21:12:38 UTC; 18s ago
       Docs: man:radiusd(8)
             man:radiusd.conf(5)
             http://wiki.freeradius.org/
             http://networkradius.com/doc/
   Main PID: 8389 (freeradius)
     Status: "Processing requests"
      Tasks: 6 (limit: 2219)
     Memory: 79.0M (limit: 2.0G)
        CPU: 314ms
     CGroup: /system.slice/freeradius.service
             └─8389 /usr/sbin/freeradius -f
ricardo@ubuntu:~$ _
```

Instalar: sudo apt install freeradius

Editar configuración: sudo nano /etc/freeradius/3.0/radiusd.conf

Syslog / rsyslog

```
ricardo@ubuntu:~$ systemctl status freeradius
● freeradius.service - FreeRADIUS multi-protocol policy server
     Loaded: loaded (/lib/systemd/system/freeradius.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 21:12:38 UTC; 18s ago
       Docs: man:radiusd(8)
             man:radiusd.conf(5)
             http://wiki.freeradius.org/
             http://networkradius.com/doc/
   Main PID: 8389 (freeradius)
     Status: "Processing requests"
      Tasks: 6 (limit: 2219)
     Memory: 79.0M (limit: 2.0G)
        CPU: 314ms
     CGroup: /system.slice/freeradius.service
             └─8389 /usr/sbin/freeradius -f
ricardo@ubuntu:~$ sudo apt install rsyslog -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
rsyslog is already the newest version (8.2112.0-2ubuntu2.2).
rsyslog set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
ricardo@ubuntu:~$ systemctl status rsyslog
● rsyslog.service - System Logging Service
     Loaded: loaded (/lib/systemd/system/rsyslog.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 20:30:33 UTC; 43min ago
TriggeredBy: ● syslog.socket
       Docs: man:rsyslogd(8)
             man:rsyslog.conf(5)
             https://www.rsyslog.com/doc/
   Main PID: 2384 (rsyslogd)
      Tasks: 4 (limit: 2219)
     Memory: 1.1M
        CPU: 53ms
     CGroup: /system.slice/rsyslog.service
             └─2384 /usr/sbin/rsyslogd -n -iNONE
ricardo@ubuntu:~$ _
```

Instalar: sudo apt install rsyslog

Editar configuración: sudo nano /etc/rsyslog.conf

SNMP

```
Get:1 http://do.archive.ubuntu.com/ubuntu jammy-updates/main amd64 snmpd amd64 5.9.1+dfsg-1ubuntu2.8
 [60.3 kB]
Fetched 60.3 kB in 1s (93.9 kB/s)
Preconfiguring packages ...
Selecting previously unselected package snmpd.
(Reading database ... 115492 files and directories currently installed.)
Preparing to unpack .../snmpd_5.9.1+dfsg-1ubuntu2.8_amd64.deb ...
Unpacking snmpd (5.9.1+dfsg-1ubuntu2.8) ...
Setting up snmpd (5.9.1+dfsg-1ubuntu2.8) ...
adduser: Warning: The home directory `/var/lib/snmp' does not belong to the user you are currently c
reating.
Created symlink /etc/systemd/system/multi-user.target.wants/snmpd.service → /lib/systemd/system/snmp
d.service.
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ricardo@ubuntu:~$ systemctl status snmpd
● snmpd.service - Simple Network Management Protocol (SNMP) Daemon.
     Loaded: loaded (/lib/systemd/system/snmpd.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2025-11-23 21:17:58 UTC; 20s ago
   Main PID: 9432 (snmpd)
      Tasks: 1 (limit: 2219)
     Memory: 4.5M
        CPU: 39ms
     CGroup: /system.slice/snmpd.service
             └─9432 /usr/sbin/snmpd -LOw -u Debian-snmp -g Debian-snmp -I -smux mteTrigger mteTrigg>
lines 1-9/9 (END)
```

Instalar: sudo apt install snmp snmpd

Editar configuración: sudo nano /etc/snmp/snmpd.conf

Nagios

```
Creating config file /etc/nagios-plugins/config/procs.cfg with new version

Creating config file /etc/nagios-plugins/config/real.cfg with new version

Creating config file /etc/nagios-plugins/config/ssh.cfg with new version

Creating config file /etc/nagios-plugins/config/tcp_udp.cfg with new version

Creating config file /etc/nagios-plugins/config/telnet.cfg with new version

Creating config file /etc/nagios-plugins/config/users.cfg with new version
Setcap for check_icmp and check_dhcp worked!
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ricardo@ubuntu:~$ systemctl status nagios-nrpe-server
● nagios-nrpe-server.service - Nagios Remote Plugin Executor
     Loaded: loaded (/lib/systemd/system/nagios-nrpe-server.service; enabled; vendor preset: enable>
     Active: active (running) since Sun 2025-11-23 21:20:57 UTC; 21s ago
       Docs: http://www.nagios.org/documentation
   Main PID: 9767 (nrpe)
      Tasks: 1 (limit: 2219)
     Memory: 1.5M
        CPU: 15ms
     CGroup: /system.slice/nagios-nrpe-server.service
             └─9767 /usr/sbin/nrpe -c /etc/nagios/nrpe.cfg -f
lines 1-10/10 (END)
```

Instalar: sudo apt install nagios4 nagios-plugins-contrib nagios-nrpe-plugin

Editar configuración: sudo nano /etc/nagios4/nagios.cfg

Fail2ban

```
Do you want to continue? [Y/n] y
Get:1 http://do.archive.ubuntu.com/ubuntu jammy/universe amd64 fail2ban all 0.11.2-6 [394 kB]
Get:2 http://do.archive.ubuntu.com/ubuntu jammy/main amd64 python3-pyinotify all 0.9.6-1.3 [24.8 kB]
Get:3 http://do.archive.ubuntu.com/ubuntu jammy/main amd64 whois amd64 5.5.13 [53.4 kB]
Fetched 473 kB in 2s (271 kB/s)
Selecting previously unselected package fail2ban.
(Reading database ... 115632 files and directories currently installed.)
Preparing to unpack .../fail2ban_0.11.2-6_all.deb ...
Unpacking fail2ban (0.11.2-6) ...
Selecting previously unselected package python3-pyinotify.
Preparing to unpack .../python3-pyinotify_0.9.6-1.3_all.deb ...
Unpacking python3-pyinotify (0.9.6-1.3) ...
Selecting previously unselected package whois.
Preparing to unpack .../whois_5.5.13_amd64.deb ...
Unpacking whois (5.5.13) ...
Setting up whois (5.5.13) ...
Setting up fail2ban (0.11.2-6) ...
Setting up python3-pyinotify (0.9.6-1.3) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ricardo@ubuntu:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/lib/systemd/system/fail2ban.service; disabled; vendor preset: enabled)
     Active: inactive (dead)
       Docs: man:fail2ban(1)
ricardo@ubuntu:~$ _
```

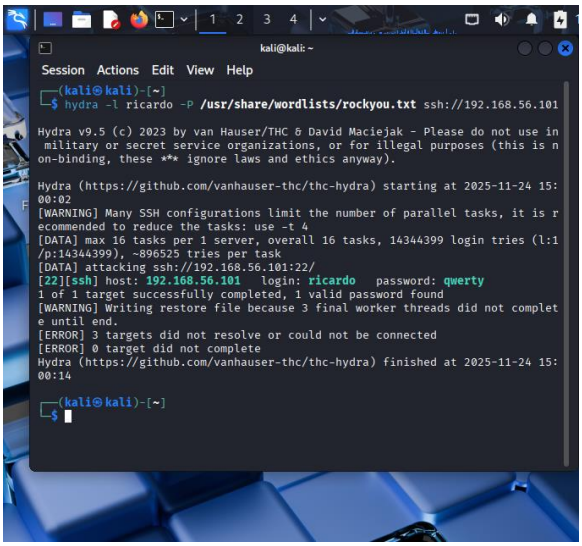
Instalar: sudo apt install fail2ban

Editar configuración: sudo nano /etc/fail2ban/jail.local

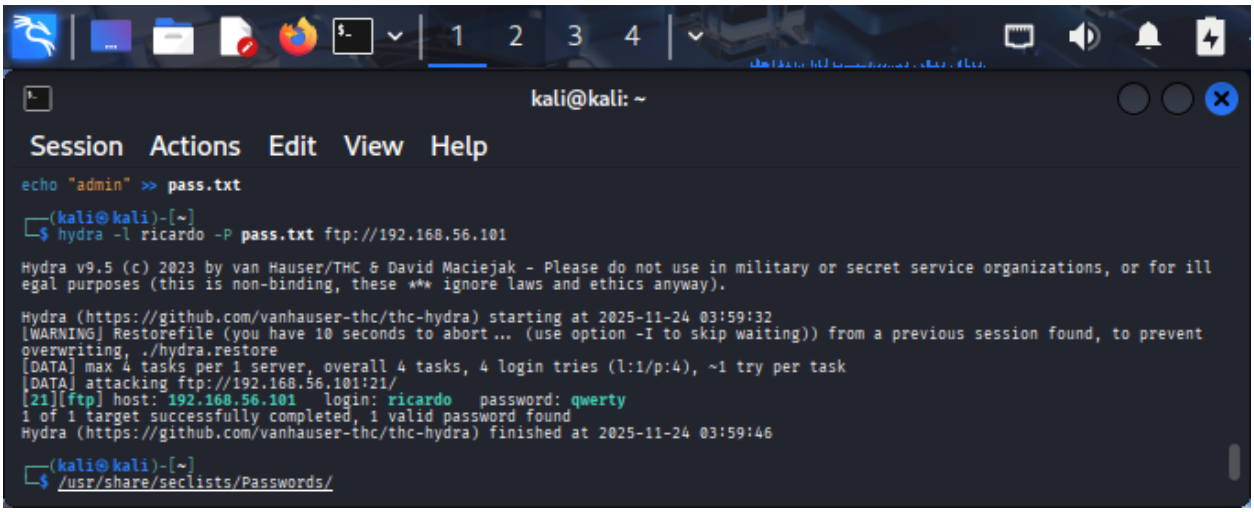
II - Evaluación de Seguridad (Sin Endurecimiento)

Empecé evaluando los puertos de la máquina víctima para identificar superficies de ataque ejecutando un escaneo con **nmap 192.168.56.101**

Con los servicios expuestos identificados, ataqué credenciales probando fuerza bruta sobre SSH y FTP usando comandos como **hydra -l ricardo -P diccionario.txt ssh://192.168.56.101** y **hydra -l ricardo -P diccionario.txt ftp://192.168.56.101**

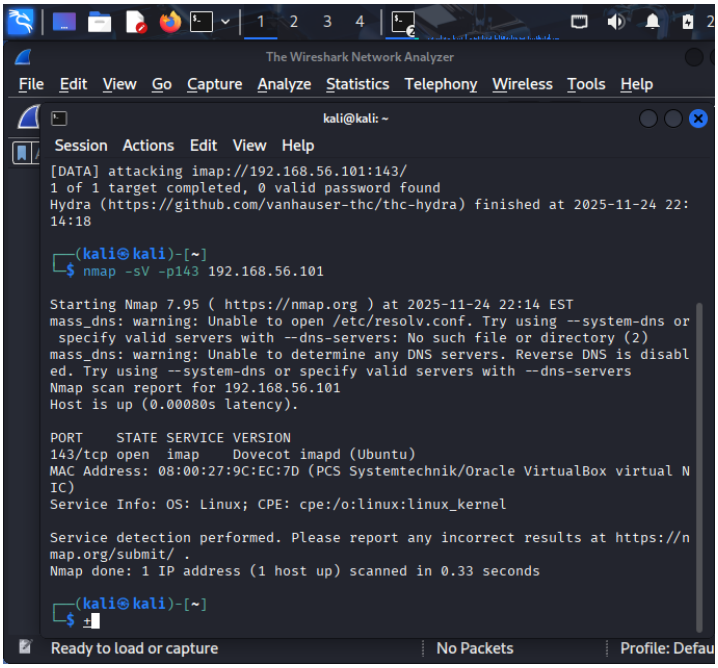


(fuerza bruta sobre SSH)



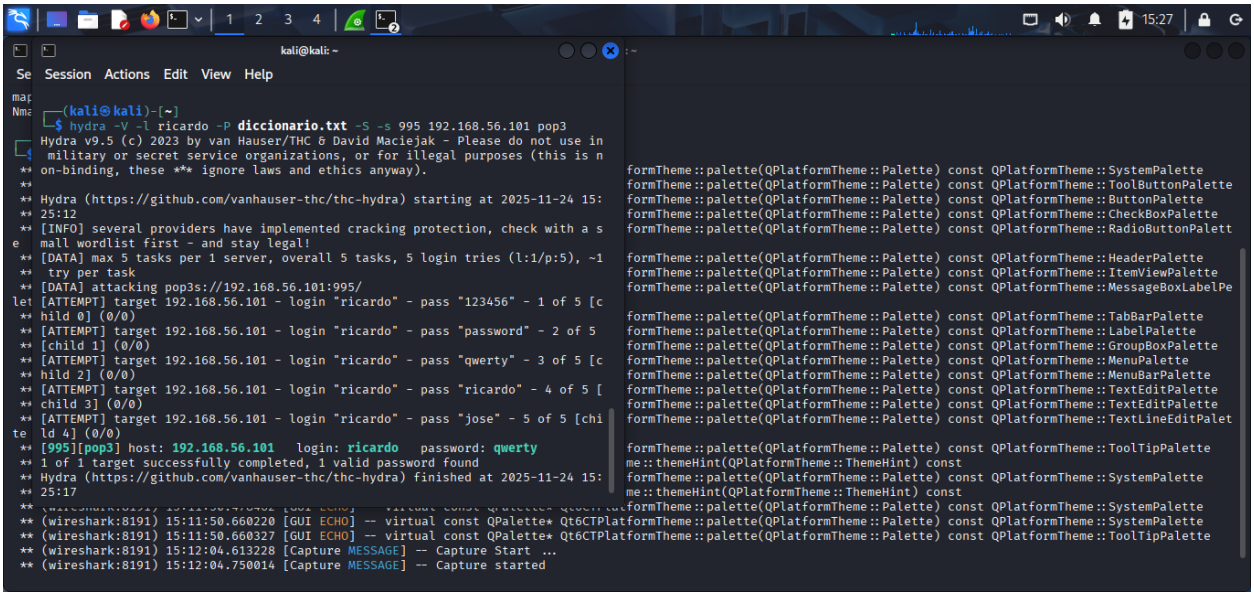
(fuerza bruta de hydra sobre FTP)

Hice un escaneo más específico con nmap a IMAP con el comando **-sV** y el puerto **-p143**:

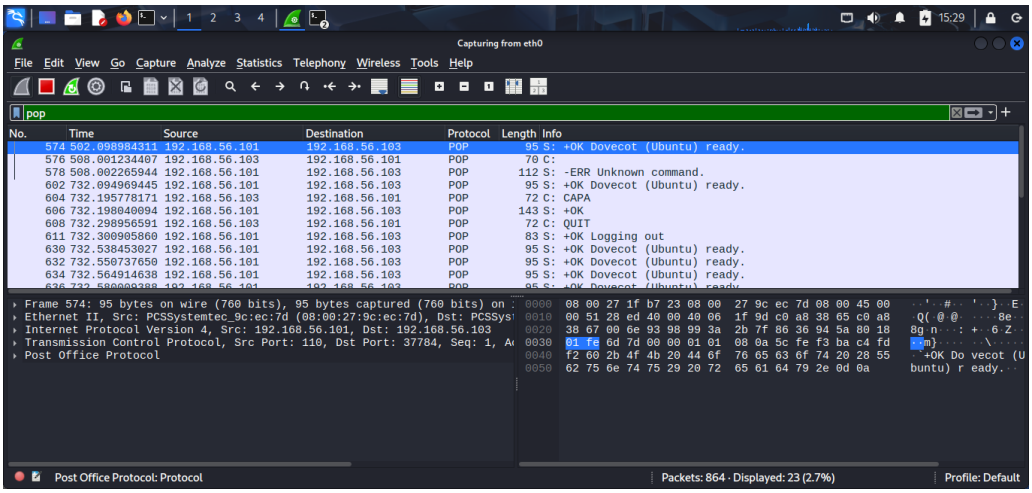


(escaneo más profundo con nmap a IMAP port == 143)

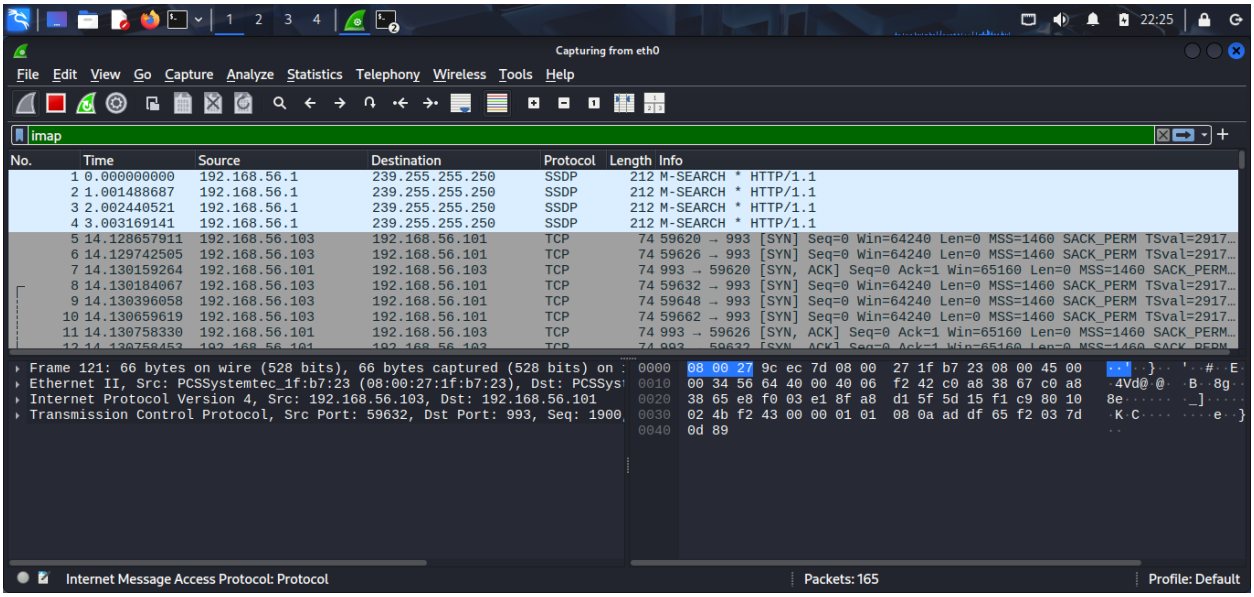
también intenté romper autenticación en POP3 e IMAP con **hydra -V -l ricardo -P diccionario.txt 192.168.56.101 pop3**



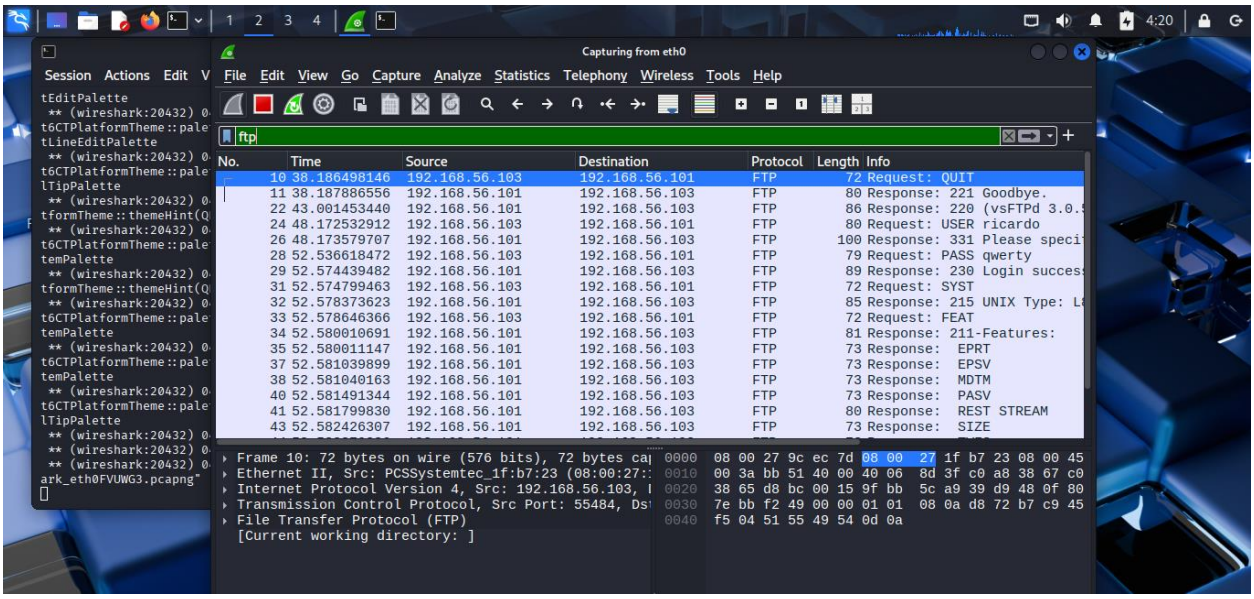
Para analizar el entorno de red y buscar credenciales en texto claro, reforcé el ataque usando Wireshark, iniciando captura en la interfaz activa y filtrando tráfico con reglas como tcp.port == 21, tcp.port == 110 o tcp.port == 143, verificando si los servicios enviaban autenticación sin cifrado.



(captura de interfaz de POP3 port == 110)

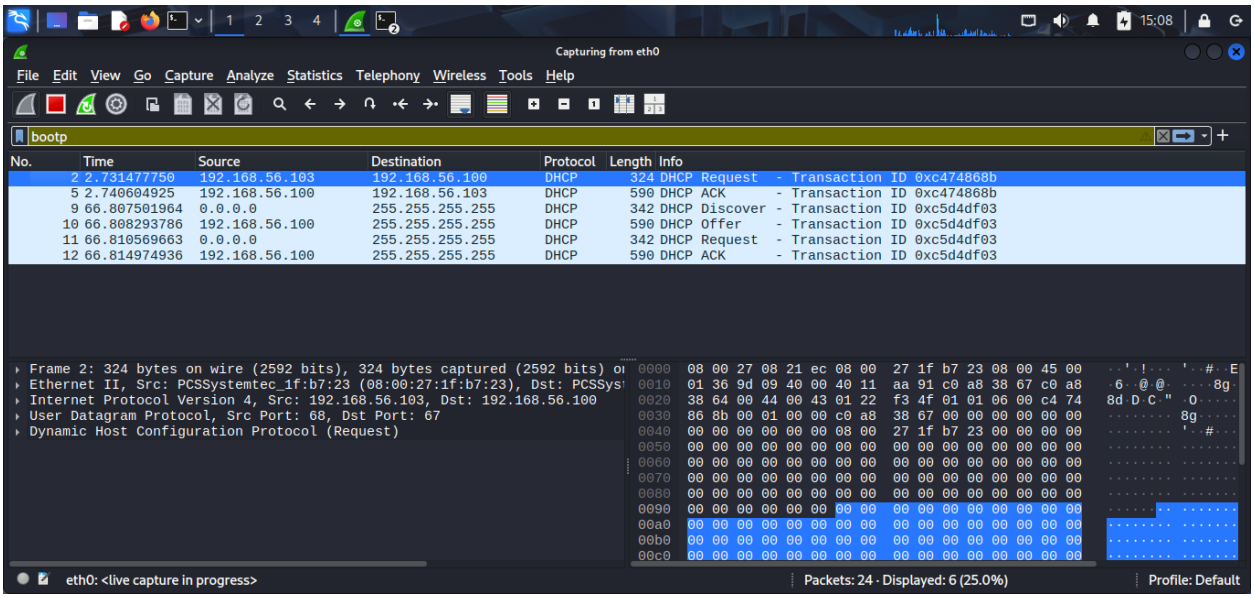


(recopilación de datos de IMAP usando wireshark port == 143)



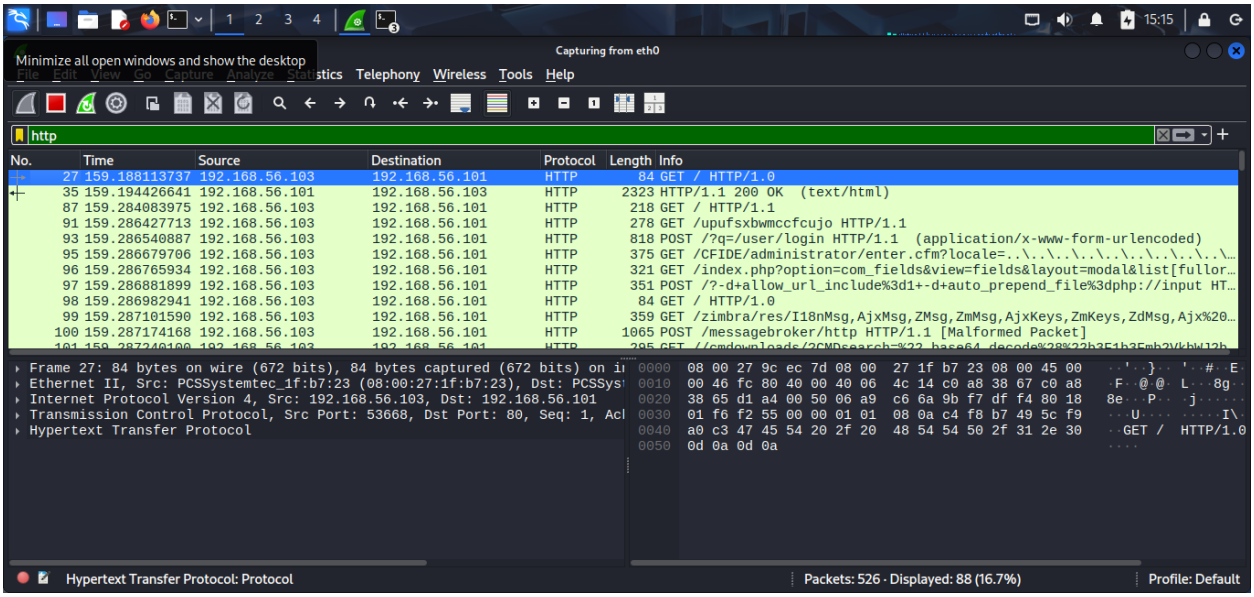
(recopilación de datos de FTP usando wireshark port == 21)

Además, comprobé servicios internos como DHCP con **systemctl status isc-dhcp-server** para ver si podía explotarse alguna mala configuración.



(recopilación de datos de FTP usando wireshark)

Para evaluar Apache, primero confirmé que el servicio web estaba activo con **nmap -p 80,443 -sV 192.168.56.101** y entrando a su página del buscador, identifiqué la versión del servidor y luego ataqué el panel de autenticación HTTP usando Hydra con **hydra -l ricardo -P diccionario.txt 192.168.56.101 http-get /**. Mientras se ejecuatab el ataque de fuerza bruta, abrí Wireshark para capturar tráfico del puerto 80 aplicando el filtro tcp.port == 80, lo que me permitió ver peticiones HTTP en texto, incluidas cosas codificadas cuando el login del sitio no tenía HTTPS. Con esto comprobé vulnerabilidad por falta de cifrado y autenticación débil.

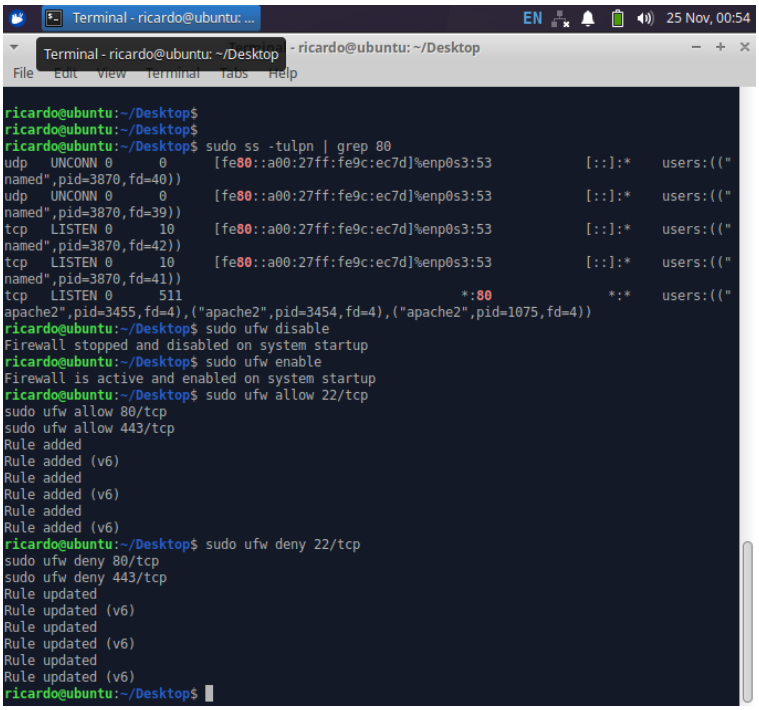


(Wireshark capturando el tráfico del puerto 80 HTTP)

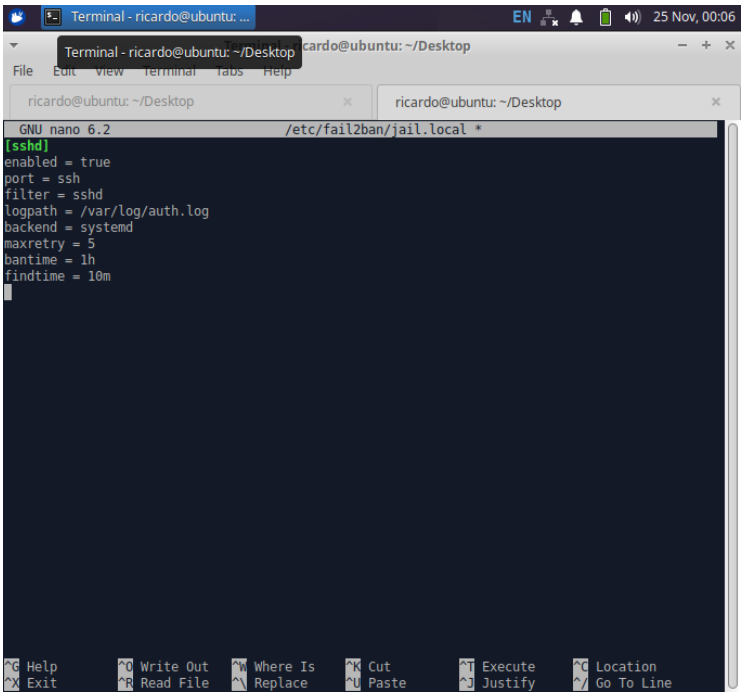
De estas fallas he analizado los riesgos principales y es que cualquier servicio sin cifrado permite robar contraseñas fácilmente, las contraseñas débiles pueden caer por fuerza bruta, y un atacante podría entrar al sistema, moverse entre servicios y escalar privilegios. Además, el tráfico capturado con Wireshark puede filtrar información sensible y la mala configuración de servicios como Apache, FTP abre la puerta a exploits o ataques.

III - Aplicación de Buenas Prácticas (Con Endurecimiento)

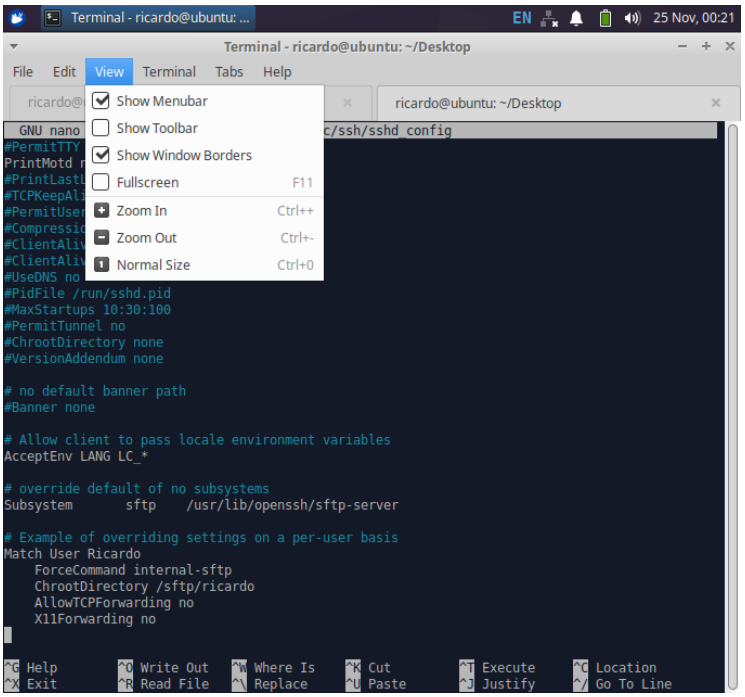
Para endurecer la seguridad de la víctima ante ataques, empecé con configurar los servicios con un enfoque más defensivo, active el firewall y el fail2ban para el control de tráfico y la protección contra ataques de fuerza bruta como hydra.



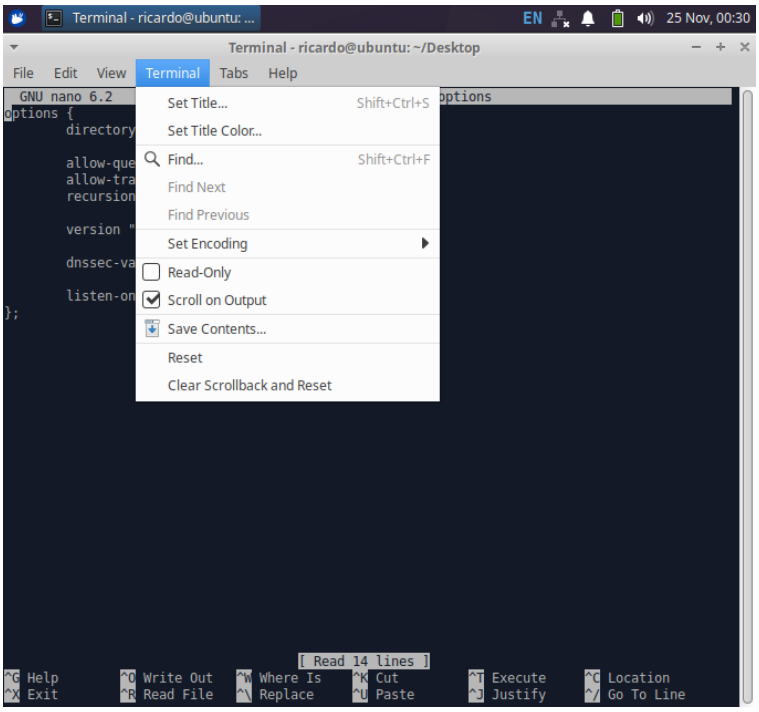
(Firewall permitiendo el tráfico a puertos específicos con el comando **allow**)



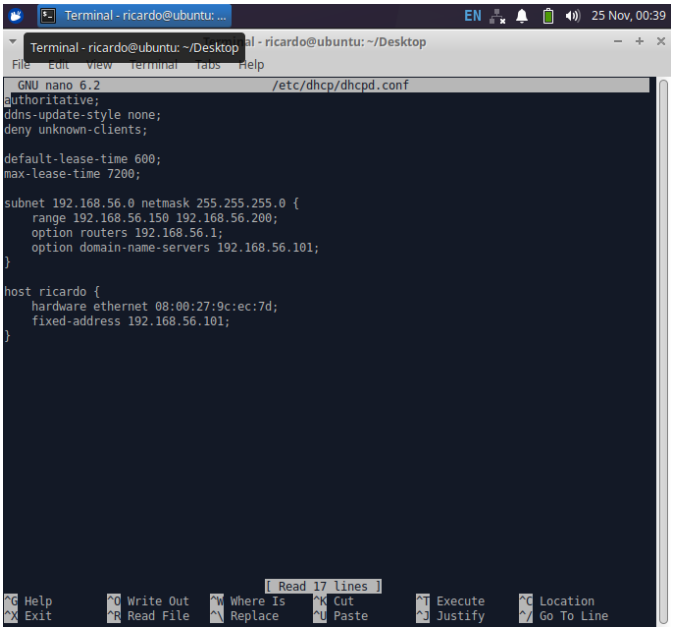
Endurecí SSH en **/etc/ssh/sshd_config** deshabilitando login de root y habilitando autenticación por llave, y verifiqué Fail2Ban con **sudo fail2ban-client status**



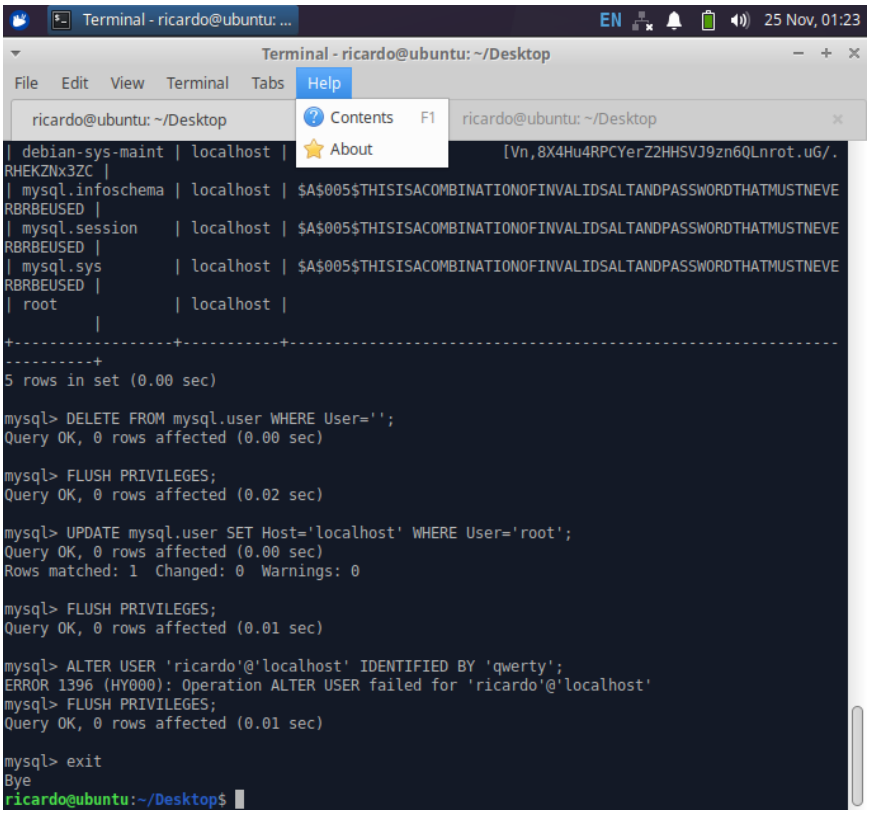
Configuré FTP/SFTP probando la conexión con sftp ricardo@192.168.56.101



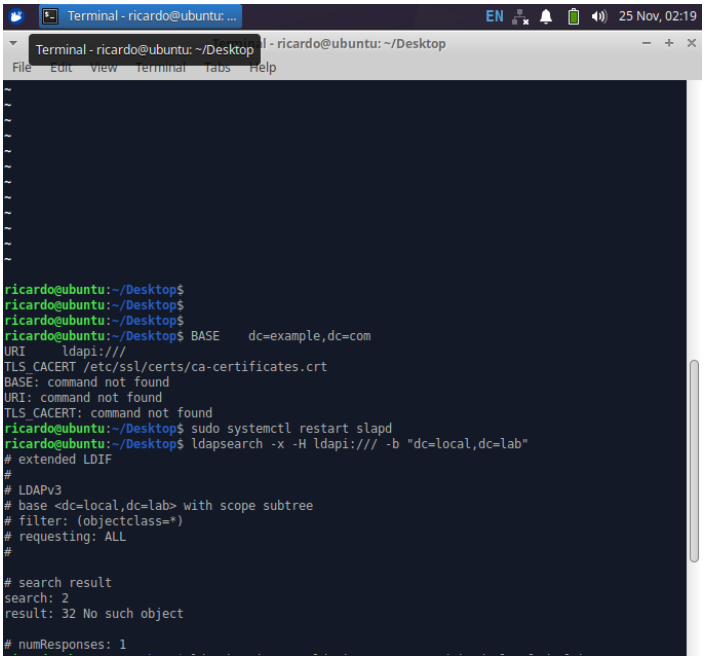
Para DNS, ajusté /etc/bind/named.conf.options escuchando solo la IP interna y habilitando DNSSEC.



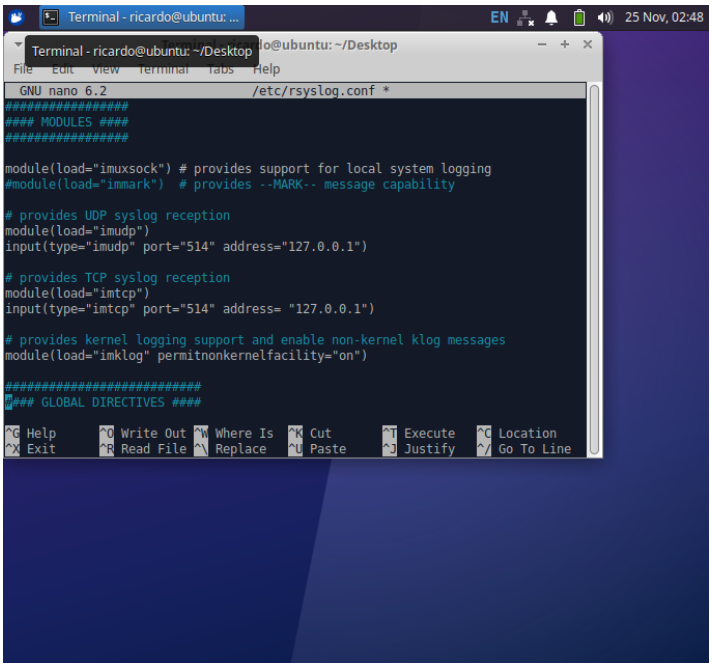
Revisé DHCP con systemctl status isc-dhcp-server y definí rangos seguros.



Endurecí MySQL deteniendo el servicio y cambiando contraseñas de usuarios con ALTER USER.



Con LDAP y Kerberos, ajusté base DN y archivos de configuración para autenticación segura.



Syslog y SNMP se configuraron para monitoreo seguro, editando **/etc/rsyslog.conf y /etc/snmp/snmpd.conf**

A lo largo de todo el proceso apliqué prácticas enfocadas a la seguridad, como: cerrar servicios innecesarios, reforzar contraseñas, habilitar firewall y Fail2Ban, limitar interfaces y puertos, y asegurar la autenticación en todos los servicios críticos. Así también como restringir el acceso de servicio luego de cierta cantidad de intentos fallidos (como en SSH)

IV - Comparativa

Cuando trabajé como atacante, busqué aumentar el área vulnerable para descubrir errores aprovechables. Usé Nmap con el fin de revisar puertos disponibles, identifiqué servicios visibles y usé Hydra para acceder mediante claves frágiles en SSH, FTP, Telnet, IMAP o también POP3. Me basé en canales no protegidos, tales como FTP, obteniendo datos sin codificar gracias a Wireshark. Esta estrategia dio resultado ya que el entorno carecía de ajustes defensivos: accesos habilitados, funciones extra operativas, autenticaciones simples, permisos ilimitados al ingresar, valores predeterminados aún activos y ningún firewall o file2ban con normas de protección.

Ahora con un enfoque más defensivo, no use servicios abiertos, busque reducir riesgos con el firewall activado y solo dejo que actué en puertos específicos que yo elija. En lugar de exponer datos importantes, las quito del sistema para que no puedan ser un objetivo fácil. Frente a claves fáciles de romper, aplico reglas más estrictas para el usuario. Además, cambio sistemas sin protección por versiones seguras con cifrado, así herramientas como Wireshark no atrapan datos sensibles. También le puse mejoras para disminuir riesgos detectados. En la tortuga, aproveché un sistema inseguro, luego lo modifiqué para hacerlo más estable.

V – Retos encontrados y las lecciones aprendidas

Lo más difícil para mí fue el aprender a usar un SO desde el inicio, así como la configuración de todos los servicios. Los problemas que tuve fueron fallos en la conexión al encender DNS y el firewall, pruebas complicadas con claves fuertes y ajustes predeterminados que chocan en servicios como Apache, DHCP o RADIUS. Varias veces al intentar configurar algo, lo “rompía” haciendo asi que dejara de funcionar por quitar una opción que no debía. Descubrí que muchos servicios traen opciones iniciales vulnerables, por eso revisar la red después del endurecimiento resulta importante para cualquier uso, ya sea profesional o personal. también tuve problemas al configurar la MV, por tener la conexión NET por defecto no podía hacer nmap desde el atacante, pero si podía hacer ping, tuve que investigar más para saber que estaba pasando pero era mas simple de lo que parecía.